

1.3 Mathematical Modeling: STFT and Mel Filterbank

► LEARNING OBJECTIVE

Rather than memorizing six decoupled DSP formulas, this section deduces each operator from first principles: the probe wave that senses harmonic resonance, the tapering window that kills boundary jumps, the Short-Time Fourier Transform that resolves the Gabor-Heisenberg uncertainty trade-off, the power spectrum that strips superfluous phase, the Mel filterbank that warps linear Hertz to cochlear biology, and the logarithmic compression that mirrors human loudness perception. Every formula is autopsied under our five-beat framework—formula, symbols, physical meaning, hardware price, and what the formula hides.

Section 1.2 constructed the datapath packaging streaming air pressure into frames. Section 1.3 is the mathematical microscope placed over a single packet, deducing each spectral operator from wave interference and auditory perception.

Six expressions transform streaming samples into 80 numbers: (1) Windowed Frame, (2) STFT, (3) Power Spectrum, (4) Mel Mapping, (5) Mel Filterbank, and (6) Log Compression. Standard parameters: $f_s = 16\,000$ Hz, frame length $= 400$ (25 ms), hop size $= 160$ (10 ms), FFT size $= 512$, filter count $= 80$, $f_{\min} = 0$ Hz, $f_{\max} = 8\,000$ Hz.

The Probe Wave, Before the Formula

How does a circuit test whether an incoming acoustic packet $\ell[n]$ vibrates at frequency f_k ? In classical physics, resonance is detected by probing: an internal reference oscillator evaluates interference against the incoming signal through an inner product $\int_0^T \ell(t) \cos(2\pi f_k t) dt$. If $\ell(t)$ contains frequency f_k in identical phase ($\cos(2\pi f_k t)$), the product becomes $\cos^2(2\pi f_k t) = \frac{1 + \cos(4\pi f_k t)}{2}$. Integrated across T , the $2f_k$ harmonic averages to zero, leaving a positive DC accumulation of $\frac{T}{2}$. If $\ell(t)$ oscillates at any different harmonic f_m ($f_m \neq f_k$), the integral evaluates to identically zero. The probe resonates exclusively with its own frequency.

Now expose the fatal flaw of a single probe wave: **phase blindness**. If the packet arrives delayed by a quarter-cycle (a phase shift of $\pi/2$, turning the input into $\sin(2\pi f_k t)$), the product becomes $\sin(2\pi f_k t) \cos(2\pi f_k t) = \frac{1}{2} \sin(4\pi f_k t)$. Over interval T , this integrates to exactly zero: $\int_0^T \sin(4\pi f_k t) dt = 0$. The single probe reports zero energy while a pure acoustic tone is roaring through the transducer.

To eliminate phase blindness, we must interrogate the wave with **two orthogonal probes in quadrature**: an in-phase probe $\cos(2\pi f_k t)$ and a quadrature probe $-\sin(2\pi f_k t)$ (90° apart). Striking this dual-branch detector, the branches measure $\cos$ and $\sin$. By Pythagorean addition, total amplitude is recovered with absolute phase immunity: $A = \sqrt{(\cos)^2 + (\sin)^2}$. Euler's identity, $e^{-j\theta} = \cos\theta - j\sin\theta$, is the algebraic packaging of this two-channel physical quadrature receiver.

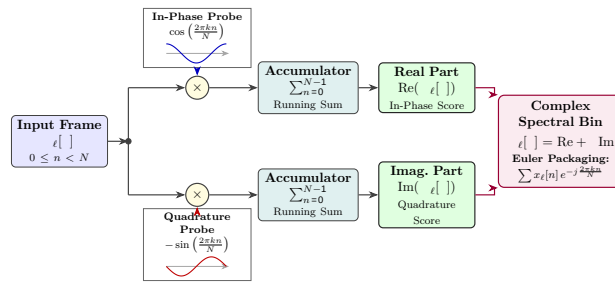


Figure 1.3a: Probe-wave quadrature interferometer. (1) **What it is:** Internal architecture of a single Fourier frequency bin evaluated as a dual-branch correlation receiver. (2) **What to look at:** Incoming frame $\ell[n]$ splits into branches multiplied by $\cos(2\pi f_k n / N)$ (in-phase) and $-\sin(2\pi f_k n / N)$ (quadrature), accumulating into real and imaginary coordinates. (3) **What it proves:** Complex numbers represent the two orthogonal axes needed to measure wave energy independently of arrival phase.

★ GROUNDBREAKING FINDING

A complex spectral bin is a dual-channel hardware quadrature receiver: two orthogonal reference oscillators ($\cos$ and $-\sin$) beating against the incoming pressure wave, returning two agreement scores that recover signal magnitude invariant to arrival phase.

1. The Windowed Frame

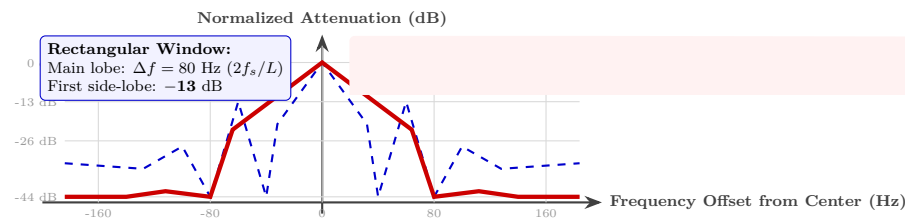
To eliminate artificial boundary step discontinuities, the raw audio packet is tapered smoothly toward zero before spectral analysis.

The 5-Beat Autopsy: The Windowed Frame

Beat 1: The Formula. $\ell[n] = \left[\frac{1}{2} + \frac{\cos\left(\frac{\pi n}{M}\right)}{2} \right] w[n]$ for $0 \leq n \leq M-1$; followed by zero-padding: $\ell[n] = 0$ for $n \geq M$.

Beat 2: The Symbols. $\ell[n]$ is the windowed frame ($M = 400$ samples, 25 ms at $f_s = 16\,000$ Hz); $x[n]$ is the streaming audio sequence; $M = 160$ samples (10 ms); $L = 400$ samples (25 ms); $w[n]$ is the fixed Hamming window, tapering boundary samples to $w[0] = w[M-1] = 0.08$ with a center peak of 1.00.

Beat 3: The Physical Meaning. Slicing rectangularly convolves the spectrum with a sinc function whose side-lobes decay slowly (first side-lobe at -13 dB). The raised-cosine Hamming window tapers boundaries to 0.08, suppressing the first side-lobe to -44 dB—a 31 dB reduction in spurious spectral leakage.



2. The Short-Time Fourier Transform (STFT)

Beat 1: The Formula. $\ell[k] = \sum_{n=0}^{N-1} \ell[n] e^{-j \frac{2\pi kn}{N}} = 0 \text{ } 1 \text{ } - 1.$

Beat 2: The Symbols. $k \in \{0 \text{ } - 1\}$ is the discrete frequency bin ($f_k = \frac{f_s}{N} = \times 31 \text{ } 25 \text{ Hz}$); $n \in \{0 \text{ } - 1\}$ is the time index; $\ell[n]$ is padded with $- = 112$ zeros to $= 512$; $\ell[k]$ is the complex spectral coefficient. By conjugate symmetry ($\ell[k -] = \ell^*[k]$), retaining the first $2 + 1 = 257$ bins (0–8 000 Hz) preserves the full spectrum.

Beat 3: The Physical Meaning. Projects the 1D temporal signal into a 2D time-frequency coordinate space, balancing the fundamental trade-off between temporal localization and frequency resolution (25 ms slices at $\Delta = 31 \text{ } 25 \text{ Hz}$).

Beat 4: The Hardware Price. Direct evaluation costs $\text{bins} \times \text{complex multiply-accumulates per } 10 \text{ ms frame}$. A radix-2 FFT exists so the designer is not stuck with 2 , reducing arithmetic to $\frac{N}{2} \log_2$ butterflies per frame. The precise slice count is Section 1.4 / Chapter 6.

Beat 5: What the Formula Does Not Say. Presumes periodic repetition over 1 . Zero-padding to 512 points interpolates the grid more densely without improving physical frequency resolution, which remains strictly limited by window length 1 .

3. The Power Spectrum

Beat 1: The Formula. $|\ell[k]|^2 = \text{Re}(\ell[k])^2 + \text{Im}(\ell[k])^2 = 0 \text{ } 1 \text{ } \frac{N}{2}.$

Beat 2: The Symbols. $|\ell[k]| \in \mathbb{R}_{\geq 0}$ is power spectral density across 257 frequency bins; $\text{Re}(\ell[k])$ $\text{Im}(\ell[k])$ are in-phase and quadrature projection scores from the STFT.

Beat 3: The Physical Meaning. Speech recognition relies on formant energy distribution rather than acoustic phase. Squaring the magnitude strips phase angle, yielding 257 non-negative real energy values.

Beat 4: The Hardware Price. The hardware price is 2 squarings and 1 addition per bin across 257 bins per 10 ms frame. Computing power ($| \ |^2$) avoids square-root hardware logic entirely. The precise slice count is Section 1.4 / Chapter 6.

Beat 5: What the Formula Does Not Say. Discarding phase is mathematically irreversible. Time-domain waveform reconstruction is no longer possible without phase estimation.

4. Mel-Frequency Mapping

Beat 1: The Formula. $\text{mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right).$

Beat 2: The Symbols. f is continuous frequency in Hertz ($0 \leq f \leq 8 \text{ } 000 \text{ Hz}$); $\text{mel}(f)$ is perceived pitch in Mels.

Beat 3: The Physical Meaning. Models human cochlear non-linearity: high sensitivity at low frequencies, coarser resolution at high frequencies. Equal steps of 355 mel span 259 Hz at the lowest band but dilate to 2 351 Hz at the highest band.

Beat 4: The Hardware Price. Zero runtime logic. The Mel warp is computed offline at design time to generate filter center frequencies stored in on-chip memory.

Beat 5: What the Formula Does Not Say. The continuous curve must map to a discrete bin grid. At low frequencies, perceptual filter bandwidths can become narrower than the digital bin spacing $\Delta = 31 \text{ } 25 \text{ Hz}$, risking discretization mismatch.

5. Mel Filterbank as a Weighted Sum

To compress linear bins into perceptual channels, triangular filters integrate energy across non-uniform bands.

The 5-Beat Autopsy: The Mel Filterbank

Beat 1: The Formula. $\ell[k] = \sum_{m=0}^{256} m[k] \cdot \ell[k] = 0 \text{ to } 256 - 1$, where $m[k]$ represents overlapping triangular weighting functions: $m[k] = \frac{k - k_{m-1}}{k_m - k_{m-1}}$ for $k_{m-1} \leq k \leq k_m$; $m[k] = \frac{k_{m+1} - k}{k_{m+1} - k_m}$ for $k_m \leq k \leq k_{m+1}$; and 0 otherwise.

Beat 2: The Symbols. $m \in \{0 \text{ to } 256 - 1\}$ is the Mel band index ($M = 80$); k is the discrete FFT bin index of the center frequency of filter m ; $m[k]$ is the triangular filter weight for bin k in filter m ; $\ell[k]$ is the integrated Mel spectral energy vector (shape $[1 \times 80]$).

Beat 3: The Physical Meaning. Eighty overlapping triangular filters compress 257 linear power bins into 80 perceptual energy channels. Each filter integrates energy across its band, preserving fine spectral detail in low-frequency vowel regions while pooling broad energy swaths in high-frequency noise regions.

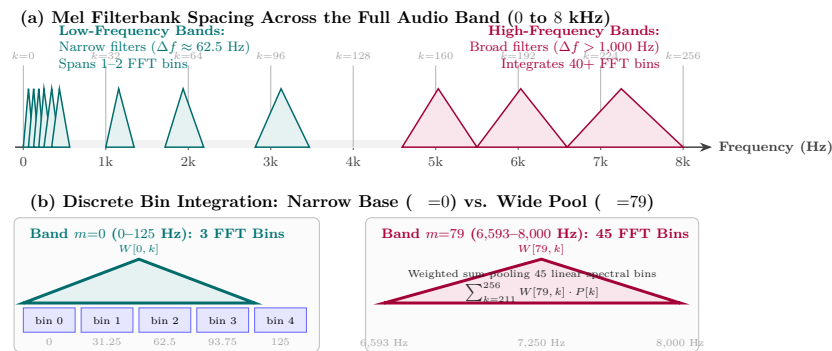


Figure 1.3b: 257 linear FFT bins collapsing under 80 Mel triangular filters. (1) **What it is:** Mapping of 257 linear frequency bins (0–8 000 Hz) into 80 perceptual Mel triangular filters. (2) **What to look at:** In region (a), low filters are dense and narrow, while high filters expand exponentially. In region (b), Band $m=0$ integrates only 3 bins, whereas Band $m=79$ pools 45 bins (6 593–8 000 Hz). (3) **What it proves:** The filterbank concentrates spectral resolution in phonetic formant bands while drastically reducing input dimensionality for downstream neural networks.

► PHYSICAL INSIGHT

The Mel filterbank is an unequal information concentrator. In low frequencies (0–1 kHz), where pitch and fundamental vowel formants reside, each band integrates only a few FFT bins, preserving fine harmonic detail. In high frequencies (6–8 kHz), where fricative consonants produce diffuse noise, each band pools dozens of bins, preserving overall energy envelope while shedding irrelevant fine structure.

Beat 4: The Hardware Price. The hardware price is a sparse matrix-vector multiplication per frame. Because each linear FFT bin falls within at most two adjacent triangular filters, the matrix is predominantly zero. The hardware price is non-zero multiply-accumulates across active bin spans; the precise slice count is Section 1.4 / Chapter 6.

Beat 5: What the Formula Does Not Say. The discretization trap. When continuous filter center frequencies are mapped to discrete bin indices, adjacent low-frequency filters can round to the identical bin if spacing is narrower than $\Delta f = 31.25$ Hz. A filter whose center and edges collapse to the same bin produces zero width and returns dead energy. Hardware filterbanks must ensure every band spans at least one unique bin.

6. Logarithmic Compression

The final transformation compresses wide physical energy variations into a normalized feature space suitable for neural network layers.

The 5-Beat Autopsy: Logarithmic Compression

Beat 1: The Formula. $\ell[\] = \ln(\max(\ell[\] 10^{-6})) = 0.1 \dots -1$.

Beat 2: The Symbols. $\ell[\]$ is the log-Mel energy feature vector (shape [1 80]) emitted to downstream acoustic models; $\ell[\]$ is the raw Mel band energy from Operator 5; 10^{-6} is the clamping floor preventing catastrophic $\ln(0) \rightarrow -\infty$ evaluation during silence.

Beat 3: The Physical Meaning. Human loudness perception scales logarithmically with acoustic power. In the physical environment, acoustic pressure spans from a quiet whisper (10^{-5} Pa) to a loud shout (10 Pa)—an energy ratio of 10^{10} , or 100 dB. Logarithmic compression normalizes this vast dynamic range into a compact numerical distribution. Furthermore, taking the logarithm converts multiplicative channel effects (such as microphone distance and room acoustics) into simple additive offsets.

Beat 4: The Hardware Price. The hardware price is 80 logarithm evaluations per 10 ms frame. In hardware, logarithms can be evaluated using lookup tables or piece-wise polynomial approximations. The precise slice count is Section 1.4 / Chapter 6.

Beat 5: What the Formula Does Not Say. The derivative of the natural logarithm, $\frac{d}{dx} \ln(\) = \frac{1}{x}$, diverges toward infinity as $x \rightarrow 0$. While clamping at 10^{-6} prevents negative infinity, tiny noise fluctuations near the clamping floor still produce large excursions in feature space, demanding careful numerical management during fixed-point quantization.

The Silicon Reality Check: Bridge to Section 1.4

The mathematical derivations of STFT and Mel filterbanks in this section assume infinite-precision arithmetic, zero-latency memory access, and instantaneous execution. On a workstation development machine, these equations run effortlessly within Python scientific environments, where abundant memory and floating-point hardware hide the 10 ms processing deadline and thermal budgets.

When this mathematical chain is pushed onto physical edge silicon operating under strict thermal limits (< 15 W) with a hard 10 ms frame deadline, abstract formulas collide with physical hardware constraints. Whether an edge GPU or a spatial FPGA fabric can satisfy these streaming acoustic deadlines is the architectural question explored next.